PT 100

Progress Log

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–We were assigned this project today and first we need to know what the project is. So we began reading the pdf that they had sent to us.

–We finally read and understood the entire pdf that they sent and it is already late for the deadline. We thought it would be better if we could first test the circuit on a breadboard so that we can make sure that there will be no mistakes when we actually solder the circuit.

–We went to the lab to receive the components required for this project and started redesigning the circuit so that it works without the help of a ADC. We went to use the Arduino analog pins for this task.

–We have just the basic circuit that they have suggested to us and we decided to go with this as we are running out of time.

–After much trouble, we have finally assembled the circuit on the breadboard. And when we turned it on, it showed the temperature as somewhere around 10 C and sometimes it would just show random numbers like -150 C.

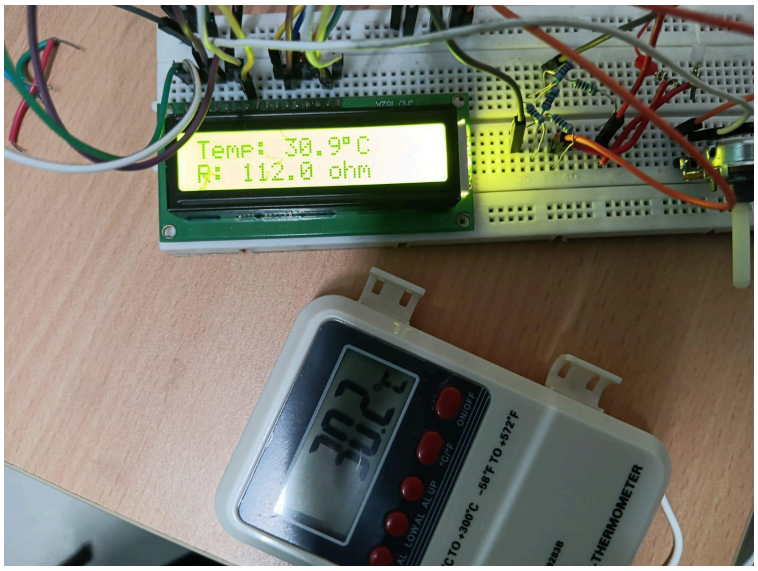
–We thought that the fluctuating values of the temperature was because of the poor connections with the breadboard and the jumper wires and also the fragile connections with the PT100. We know that when we solder it, such fluctuations will not arise.

–Now we thought that these incorrect value of temperature are due to the calibration values provided by the pdf as these values are component specific. We first measured the room temperature and used AI to give us the variables, which needs to be shifted.

–We found that the 120 ohm resistor that we originally used actually has 107.5 ohm resistance. This makes a huge difference in the accuracy of the temperature as the even small changes in the resistance will be exaggerated to about 1 C.

–We recorded the values of the temperature at various temperatures with a mug of hot water and a reliable thermometer. We gave these values to AI to generate the accurate coefficients for the temperature measurement.

–After much trouble, we got temperature reading much close to the actual temperature. We are able to achieve accuracy of about .7 C. Now, that we have a working thermometer, We decided to finally solder the circuit as it is.



–We were already running late and the soldering machines were fully busy and we had a hard time finding a solder machine and started soldering.

–The soldering was a lot harder due to the fact that all the soldering machine tips were really big and also the machines have only certain hotspots , which have the capability of melting the solder. And also we do not have any flux and the solder is getting oxidized very fast and messing with the soldering machine. Also the pins of the LCD display are very closely spaced together that very much often the pins accidentally solder to the pin beside it.

–We soldered other parts which were relative easy to solder as they are placed far apart. But the LCD is giving us so much trouble that both of us got really tired of it. After spending so much of time on the soldering, we finally did complete the soldering and then we connected everything to the arduino.

–When we turned it on with the same code, it initially displayed some random value and we really felt bad after doing much work. But for some reason when we tried it again after some time, it actually displayed the temperature. But the thermometer need some more calibration as the contact resistances changed.

–We made the 3D files which took a lot of time than expected. And we decided to use nails to coles the box .

–We started caliibrating the thermometer so that it is accurate to the actual temperature.